



Sparse Dictionary Learning for Interpretable Lung Disease Diagnosis from 3D Chest CT

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Abstract

Abstract goes here.

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Acronyms

AI Artificial Intelligence

C/G Consolidation-to-Ground-Glass-Opacity

CNN Convolutional Neural Network

CT Computed Tomography

GGO Ground-Glass Opacity

HU Hounsfield Unit

KSVD K-Singular Value Decomposition

LC-KSVD Label-Consistent K-Singular Value Decomposition

Chapter 1

Introduction

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality. In the diagnosis of lung disease, assessing the pattern, location, and regional distribution of involvement is the domain of radiology. Clinical diagnosis increasingly relies on radiological patterns to differentiate between pathologies with overlapping symptoms. While Chest X-rays (CXR) remain the first-line screening tool due to low cost and radiation, they suffer from limited sensitivity (roughly 70% visibility of lung volume) due to anatomical superposition, (where all anatomical structures located in the path of the X-rays are stacked onto a 2D plane). [1]

CT is a radiographic technology that combines X-ray equipment with a computer and a cathode ray tube display to produce images of cross sections of the human body. [2] ^{**}(Advantage of CT vs X-ray). The fundamental metric of a CT scan is the Hounsfield Unit (HU), a relative quantitative scale that describes radiodensity [1.1] In pulmonary imaging, HU values allow for the objective differentiation of tissues that might appear visually similar.

A key characteristic in CT is "Ground-Glass Opacity" (GGO), which appears as hazy increased opacity of lung, with preservation of bronchial and vascular margins (Fig 1). "Consolidation" on the other hand, appears as a homogeneous increase in pulmonary parenchymal attenuation that obscures the margins of vessels and airway walls (45) (Fig 2). The attenuation characteristics of consolidated lung are only rarely helpful in differential diagnosis. [46] Consolidation is more opaque than GGO. While GGO often suggests a reversible or early-stage process, consolidation usually indicates a more aggressive or advanced stage. The Consolidation-to-GGO (C/G) ratio is a quantitative metric calculated by dividing the volume or area of consolidation by the volume or area of ground-glass opacity within a lesion or throughout the lungs.

The traditional approach for CT analysis relies on a "slice-by-slice" approach where a radiologist views a series of axial cross-sections and identifies the "index slice"—the single image where a pathology appears at its largest diameter. A 2D slice only captures a small fraction of a 3D structure. If a nodule is irregularly shaped, the "largest" slice may not represent the true extent of the disease [2.2]. Studies show that 2D measurements are highly sensitive to the patient's orientation and the specific slice chosen, leading to high inter-observer variability [2.3]. 2D models treat each slice as an independent data point, losing the "spa-

tial continuity” between adjacent tissues [1.4]. 3D volumetric analysis represents a quantitative shift wherein the entire dataset is used to reconstruct the lung’s anatomy as a solid volume. 3D advantages?? [5.1].

Modern radiology increasingly relies on Artificial Intelligence (AI) to streamline the acquisition of radiologist analyses on chest X-rays, reinforcing the potency of automated triaging systems in streamlining healthcare workflows and amplifying patient care standards [42][1]. However, the exponential increase in data volume generated by 3D CT scanners presents a profound challenge for human interpretation and computational analysis. While the prevailing research trend has focused heavily on the deployment of 3D deep learning architectures, such as Convolutional Neural Networks (CNNs), these models frequently encounter significant hurdles regarding computational overhead, data scarcity, and a critical lack of interpretability. The research proposed herein advocates for a departure from these ”black box” paradigms, suggesting that sparse representation methods offer a superior framework for the volumetric identification of lung diseases by ensuring clinical interpretability.

1.2 Problem Statement

Modern radiology increasingly relies on AI to manage the vast quantities of 3D data produced during routine screenings. 3D deep learning has been widely applied to the automatic diagnosis of lung diseases, achieving remarkable success in tasks such as segmenting suspected infection regions and differentiating between various types of carcinoma. Despite these advancements, the inherent complexity of 3D CNNs introduces a set of constraints that are often overlooked in the pursuit of high accuracy scores.

1.2.1 Computational Overhead

Processing a 3D CT scan involves treating stacked 2D slices as a single volumetric matrix, which demands significantly more GPU memory and processing power than standard image analysis. Unlike ordinary RGB images, CT scans consist of single-channel data with varying voxel dimensions and resolution, which complicates spatial alignment and generalizability across different scanners. Furthermore, the ”curse of dimensionality” remains a persistent threat; as the network depth increases to capture global features, the risk of gradient explosion and slow convergence becomes more pronounced.

1.2.2 Interpretability Bottleneck

Beyond computational limits, the clinical utility of deep learning is hampered by the interpretability paradox. While tools like Grad-CAM provide visual heatmaps to localize lesions, these post-hoc explanations are often insufficient for medical professionals who require a granular understanding of the model’s decision-making process. In high-stakes environments, a model that shruggingly produces a diagnosis without an audit trail of its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance.

This transparency gap has led to a renewed interest in "open book" models that are interpretable by design, where the internal logic is clear from the outset.

1.3 Objectives and Scope

1.3.1 Objectives

1. Develop a Dictionary Learning framework for Interpretable Lung Disease Classification from 3D Chest CT
 - Implement training in two phases for normal and pathological atoms
 - Integrate KSVD with label consistency constraints
2. Visualize dictionary atoms and map them to anatomical features
 - Compare sparse coefficient patterns across disease classes
 - Demonstrate atom-to-pathology correspondence
3. Benchmark against baseline models
 - Compare with standard KSVD without label consistency constraint
 - Compare with 3D CNN baseline (3D-ResNet)
4. UI ??
5. Integrate clinical metrics for volumetric analysis
 - Implement HU-based segmentation for GGO and consolidation
 - Calculate C/G ratio and compare model predictions against clinical thresholds

1.3.2 Scope

The proposed solution will only target Chest-CT scans and will be limited to pulmonary diseases, particularly ones labelled in the utilized datasets.

Chapter 2

Literature Review

Lung diseases are one of the biggest health problems in the world today, causing a lot of illness and death. The Global Burden of Disease Study indicates that chronic respiratory conditions, including chronic obstructive pulmonary disease (COPD), asthma, interstitial lung disease, and lung cancer, resulted in about 4.2 million deaths and impacted approximately 569 million people worldwide in 2023 (GBD Chronic Respiratory Disease Collaborators, 2025). Lower respiratory infections, especially pneumonia, remain the seventh leading cause of death worldwide, disproportionately affecting children under five and elderly populations in resource-poor areas (Troeger et al.). In the year 2025. The outbreak of the COVID-19 pandemic highlighted the continued vulnerability of respiratory health, while simultaneously accelerating advancements in diagnostic imaging and AI-based detection methods (WHO, 2024).

Traditionally, doctors have used a blend of clinical evaluation, pulmonary function tests, and imaging studies to identify respiratory ailments. CT scans are considered the best way to look at the lungs in detail. High-resolution CT and 3D CT scans can show small changes in the lungs, like ground-glass opacities, nodules, fibrosis, and areas of consolidation. But reading these images is time-consuming and can vary between different doctors. With more images being taken every day, this problem is getting bigger. That’s why researchers are developing tools like computer-aided diagnosis systems that use machine learning and deep learning to help doctors make better and faster diagnoses.

In the field of AI for medical imaging, there’s a big challenge between how well the AI can predict and how clear its reasoning is.

Deep learning models like CNNs and vision transformers have done very well in classifying lung diseases, sometimes even better than experts. But these models often act like “black boxes,” giving answers without explaining their reasoning. This makes it hard for doctors to trust and use them in real medical situations. To solve this, researchers are looking into explainable AI and other methods like sparse dictionary learning that can offer clearer, more understandable results that match what doctors see in the body.

This literature review offers a detailed examination of the existing research works that supports to the proposed study, which involves the use of sparse dictionary learning for interpreting lung diseases through 3D chest CT scans easier to understand. it starts with an introduction to deep learning methods for detecting lung diseases, then proceeds to sparse coding and dictionary learning techniques,

followed by radiomics and CAD systems, explainable AI in medical imaging, and recent developments in transformer-based architectures. The chapter ends with a look at what’s still missing in this area and what the new research plans to fix.

2.1 Previous Work

2.1.1 Evolution of CNN-Based Approaches

The application of deep learning to lung disease diagnosis has witnessed exponential growth over the past decade. Deep learning architectures, specifically Convolutional Neural Networks (CNNs), have become the leading approach for automated analysis of chest X-rays and CT scans. Recent studies show that deep learning models surpass traditional machine learning methods by achieving classification accuracy above 98% for multi-disease detection tasks [1][2].

Modern studies are increasingly concentrating on 3D convolutional neural networks that view CT scans as complete three-dimensional data rather than separate two-dimensional slices. This method, which focuses on volume, better identifies spatial connections within lung tissue, thereby enhancing the ability to detect minute pathological alterations. Li et al. In 2025, it was shown that preprocessing methods like lung region extraction, combined with advanced backbone architectures like ResNeSt50, markedly improved classification performance for multiple lung pathologies, including adenocarcinoma, squamous cell carcinoma, and benign nodules [3][4].

Employing pre-trained models such as DenseNet-201, VGG16, and ResNet for transfer learning has shown significant efficacy when dealing with limited annotated medical imaging datasets. Combining multiple pre-trained models through ensemble techniques typically demonstrates superior performance over single-model approaches, particularly in multi-class classification tasks involving diverse lung conditions such as pneumonia, COVID-19, emphysema, and lung cancer [5][6].

2.1.2 Sparse Dictionary Learning in Medical Image Analysis

Sparse dictionary learning represents an unsupervised machine learning method for extracting interpretable, meaningful features from medical images. This technique learns a set of “dictionary” elements or basis vectors that represent data in a sparse manner. Therefore each image or patch is modeled as a linear combination of a small subset of learned dictionary atoms [7].

This framework highlights multiple attractive features for medical imaging. The well-learned dictionary atoms frequently relate to understandable local features such as edges, textures, or anatomical structures, rather than the abstract, distributed representations commonly found in neural networks. The sparse representation of the data offers a form of regularization that enhances generalization from limited training data, that benefits when dealing with limited annotated medical data.

Sparse coding has been utilized in various medical image analysis tasks. Sparse

priors in image reconstruction facilitate high-quality reconstruction from under-sampled data, thereby speeding up MRI scans or decreasing CT radiation exposure. In segmentation tasks, dictionary-based methods excel at identifying distinctive tissue features, which aids in the automated identification of anatomical structures or abnormalities (Ravishankar et al.). In 2020.

Studies with classification tools have shown that minimal feature sets can efficiently capture pathological characteristics. Sparse discriminative learning in brain imaging has facilitated precise distinction between healthy and pathological tissue conditions. In the context of chest imaging, researchers have investigated sparse representations for detecting tuberculosis, characterizing lung nodules, and classifying interstitial lung diseases..etc (Zhang et al.). In 2018; Song et al. In 2015.

Recent studies are investigating hybrid methods that integrate sparse coding with deep learning techniques. Sparse autoencoders, for example, introduce sparsity constraints on neural network activations, promoting disentangled representations where individual neurons correspond to meaningful features. Research indicates that these models are capable of producing radiology reports that match expert interpretations while ensuring transparency regarding the specific features of images that lead to particular conclusions (Chen et al.). In 2024.

The main benefit of sparse dictionary learning for clinical use is its inherent ease of interpretation. When dictionary atoms relate to identifiable anatomical or disease-related features, healthcare professionals can directly assess which patterns are responsible for a particular classification. This contrasts with post hoc explanation techniques used with black box models, which aim to replicate decision-making logic after the fact rather than integrating transparency into the model’s design.

2.1.3 Explainable AI in Medical Imaging

The integration of AI tools in medical imaging has encourage extensive studies on explainable artificial intelligence (XAI). The “black-box” characteristic of deep neural networks poses substantial hurdles for clinical implementation, necessitating physicians’ comprehension of diagnostic reasoning for accountability, patient safety, and optimal care delivery. [9][10].

Common XAI techniques in medical imaging, like:

- Visual explanation methods, use methods such as Grad-CAM and Integrated Gradients to create saliency maps that show which parts of the images are most crucial for model predictions [11].
- Feature attribution methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) offer quantitative insights into feature contributions [12].
- Attention mechanisms that allow models to selectively focus on relevant image areas, providing clearer decision-making logic [13].

Research indicates that these techniques enhance clinicians’ comprehension and confidence in AI-generated diagnostic results, enabling seamless incorporation into conventional radiology practices while maintaining regulation. [14].

2.2 Gaps in Literature

2.2.1 Integration of interpretability with 3D volumetric analysis is limited.

Despite the significant advancements in 3D Convolutional Neural Networks and volumetric analysis techniques for lung disease classification from CT scans, most explainability studies have concentrated on 2D imaging modalities like chest X-rays or individual CT slices analyzed separately. A substantial disparity exists in creating interpretable feature representations tailored for comprehensive 3D volumetric data.

Current post-hoc techniques for analyzing 3D data (such as 3D Grad-CAM) offer visualizations of significant areas, yet they lack semantically meaningful feature representations that clinicians can easily connect to anatomical structures or pathological patterns. The application of 3D saliency maps to understand specific lesion types, textural patterns, or anatomical landmarks is still largely unexplored. This particular gap is notably important because pulmonary pathology frequently appears as spatial patterns across multiple CT slices, such as centrilobular versus panlobular emphysema distribution.

2.2.2 Relying on Post-hoc Explanations Over Inherent Interpretability.

Modern XAI methodologies for medical imaging frequently utilize post hoc explanations for opaque deep learning models. These approaches explain actions taken afterward but fail to tackle the black-box issue—underlying models remain intricate and opaque, regardless of accompanying explanations.

Sparse dictionary learning presents an alternative approach where learned dictionary elements can represent interpretable features such as nodules, ground-glass opacities, consolidations, and honeycombing. Every classification decision links to particular, tangible dictionary components, offering inherent rather than post-implementation clarity. Despite the integration of sparse dictionary learning principles with contemporary 3D deep learning models for analyzing lung CT scans, significant exploration in this area is still lacking. The existing hybrid methods, such as deep dictionary learning, have mainly been applied to 2D natural images rather than to volumetric medical imaging.

2.2.3 The Accuracy-Interpretability Trade-off

A significant obstacle in medical AI development lies in the tension between achieving high accuracy in models and maintaining interpretability. More straightforward, easier-to-understand models like decision trees, sparse linear models, and classical dictionary learning might fall short in diagnostic accuracy when compared to deep neural networks with hundreds of thousands of parameters. Previous research fails to provide unified approaches that balance both high classification accuracy and interpretability in 3D lung imaging studies.

The task of creating architectures that match the accuracy of current deep models but offer interpretable representations aligned with clinical knowledge con-

tinues to be an open research issue. Efficient dictionary learning, which represents data using learned basis elements, presents a viable solution for bridging this gap. However, its scalability to 3D volumetric analysis necessitates innovative methodologies.

2.2.4 Clinical Validation of Learned Representations

Numerous studies suggest interpretable AI techniques and visualize learned features, but clinical validation showing that these representations align with patterns recognized and trusted by pulmonologists and radiologists remains limited. The discrepancy between computational interpretability metrics, such as faithfulness scores and attention map entropy, and actual clinical effectiveness persists.

Research examining whether domain experts perceive learned representations as meaningful, whether explanations enhance diagnostic confidence, and whether interpretable models facilitate clinical workflow integration is notably scarce. The field demands uniform for examining how humans interact with AI in studies that evaluate the practical significance of interpretability in clinical decision-making scenarios.

2.2.5 Handling Class Imbalance with Interpretable Models

Uncommon lung conditions such as large cell carcinomas, carcinoid tumors, and rare interstitial lung disease patterns like pleuroparenchymal fibroelastosis, along with emerging infectious diseases, pose significant class imbalance issues. Deep learning studies have tackled this issue using weighted loss functions, oversampling techniques, synthetic data creation (such as progressive growing GANs), and meta-learning methods.

Despite limited exploration, integrating class imbalance mitigation strategies into interpretable sparse representations for 3D CT analysis remains largely uncharted. Dictionary learning methods that efficiently identify representative atoms for rare categories—while preserving interpretability for common diseases—require further improvement. Although sparse dictionary-based few-shot learning approaches show promise for rare disease detection, their effectiveness in pulmonary imaging has not been thoroughly investigated.

2.2.6 Standardized Frameworks for Evaluating Interpretability

The field is short of uniform standards for measuring the interpretability of lung disease diagnosis systems. Recent research utilizes diverse metrics, datasets, disease classifications, and evaluation techniques, which complicate the direct comparison of interpretable methods and hinder clinical application.

Using standardized benchmark datasets with expert annotations of clinically significant features, rather than just disease labels, would aid in the creation and evaluation of interpretable models. Essential protocols for assessing both predictive accuracy and explanation quality, validated by clinician feedback, are required to advance the field towards clinical implementation.

2.2.7 Sparse Dictionary Learning for Three-Dimensional Medical Imaging.

Advanced sparse coding and dictionary learning techniques have been mainly devised for 2D image patches, with adaptations needed for 3D data. Convolutional sparse coding offers an additional application, yet it is most commonly utilized in natural videos and microscopy, rather than in volumetric medical imaging.

The advancement of fast computational methods for creating 3D dictionary models from CT scans, coupled with their seamless incorporation into deep learning systems for complete optimization, remains an open research issue. Techniques that capture clinically relevant 3D patterns through specialized dictionaries, avoiding generic textural features, and ensuring computational efficiency for clinical applications must be explored further.

Chapter 3

Methodology

The research consists of main phases Overall methodology Diagram

3.1 Literature Review

following topics are covered in literature review existing solutions and their gaps research frontiers

3.2 Data acquisition and pre-processing

We selected two publicly available chest CT datasets, CT-RATE[citation] and ReXGroundingCT[citation], for training and evaluation. due to their complementary nature. CT-RATE is an open-source dataset of 50,188 reconstructed 3D chest CT volumes with their corresponding radiology reports. Each volume is annotated with 18 abnormalities, supporting extensive validation and fine-tuning. ReXGroundingCT extends CT RATE by providing expert annotated 3D segmentation masks that highlight the exact image regions referenced in the radiology reports. The dataset contains 3,142 chest CT scans across 14 abnormality categories. Both datasets are released under MIT licenses that allow their use for non-commercial academic research.

Data pre-procesising corrections such as intensity normalization and spacing adjustments have already been implemented directly in the provided NIfTI files. If further pre-processing is required, we will apply standard techniques such as resampling to a uniform voxel size, intensity normalization, and lung segmentation, before feeding the data into our model.

3.3 Model Training

Model training will be implemented using a two-phase frozen dictionary learning approach to capture both normal and pathological patterns in chest CT scans. The dataset will be divided into train,test, and validation sets.

Phase 1: Learning the Normal Dictionary We will first train a dictionary using only healthy or normal samples from the dataset. This step will allow the system to learn the typical structures of the lungs

Phase 2: Learning Pathological Atoms After the normal dictionary is established and fixed, we will train additional dictionary atoms to capture disease-specific or pathological features. While the normal dictionary remains frozen, it will contribute to the sparse coding process, allowing the model to efficiently learn only the new pathological features.

We extend the dictionary learning approach using LC-KSVD to improve the discriminative power of the learned sparse representations. The label consistency constraint will produce similar sparse codes for CT samples belonging to the same disease category.

We will formulate LC-KSVD to jointly optimize three key objectives:

1. Accurate reconstruction of CT image patches using the learned dictionary.
2. Consistency of sparse representations within the same disease class.
3. Effective classification of lung conditions based on the sparse codes.

Appropriate weighting will be applied to balance these objectives, and hyperparameters will be selected through validation experiments to ensure stable and meaningful learning.

3.4 Classification and Disease Diagnosis

Support Vector Machine (SVM) use as the final disease classification model because of several features: 1. Ability to handle high-dimensional sparse features produced by dictionary learning and LC-KSVD 2. Effectiveness when the number of features is much larger than the number of samples 3. Strong generalization due to the maximum-margin principle

SVMs are also known to be robust when dealing with limited and imbalanced medical data, making them a suitable choice for disease classification tasks in chest CT analysis.

The model will be implemented using the scikit-learn library in Python and will receive the sparse codes generated from the learned dictionary as input features. A linear kernel will be used as the initial choice due to its effectiveness with sparse representations.

The classifier will predict multiple disease categories, with the number of output classes matching the labeling structure of the CT-RATE dataset.

3.5 Interpretability Analysis and Visualization

To improve interpretability and clinical relevance, the learned dictionary atoms will be visualized and analyzed through an intuitive UI/UX design that incorporates 3D rendering techniques. Each dictionary atom will be mapped back to the original CT image space, allowing us to see which anatomical regions or pathological patterns the model has learned to represent. This mapping is guided by the ReXGroundingCT annotations, which provide precise spatial localization of abnormalities within the 3D imaging volume.

3.6 Volumetric Feature Extraction and Quantification

Ground-glass opacities (GGO) will be identified based on their characteristic intensity patterns in CT images, measured using Hounsfield Units (HU). GGO regions are defined as lung areas with attenuation values then captures the partially aerated lung tissue associated with GGOs while excluding normal lung parenchyma and denser consolidations.

3.7 Baseline Comparisons and Benchmarking

To demonstrate the effectiveness of the proposed method, it will be compared against well established baseline approaches commonly used in current medical imaging research. These baselines are chosen to represent both traditional machine learning techniques and modern deep learning models. Standard K-SVD is included as a baseline to highlight the advantage of incorporating label information through LC-KSVD. By comparing these two sparse representation methods, the contribution of label consistent dictionary learning to disease classification performance can be assessed. In addition, several 3D CNN architectures will be used as deep learning baselines, as they are considered state-of-the-art for volumetric CT analysis. These models are capable of learning complex spatial patterns across multiple slices and therefore provide a strong benchmark against which the proposed method can be evaluated. Existing implementations of these networks will be adopted to ensure reliability and consistency. All models will be trained and tested using the same dataset, although training strategies may differ due to the higher data and computational demands of CNN-based approaches.

Chapter 4

Timeline and Resource Required

4.1 Timeline

Provide a realistic timeline for completing the project. Use a Gantt chart or table.

4.2 Resource Required

State the resource required for the project and estimate the budget for the resources required.

Chapter 5

Conclusion

Summarize the key points of your proposal and reiterate the importance of the project.

References